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Título/Title

Program PROCESASEC, Version 2
User's Manual

Palabras clave/Key words	Clasificación acceso Access class **P1**
MANUALS / STRESS ANALYSIS / COMPUTER PROGRAMS	Registro de Revisiones Revision Record **Pag. 2**

Resumen/Summary

The purpose of this Technical Report is to define the way to use the PROCESASEC program.

This program processes a flight-by-flight, cycle-by-cycle random stress sequence. The input sequence can be provided in two different ways:

- Sequence of discrete flights.
- Continuous stress recording.

The processed sequence will be generated with the following particularities:

- The stress sequence is filtered, removing all intermediate stress states.
- The sequence can be modified by applying a factor on stresses and adding a constant stress level.
- Small cycles can be removed from the final sequence by defining an appropriate omission level.
- A Rainflow counting method can be applied on the sequence.
- A block spectrum is also generated from the final stress sequence.

The final stress sequence is written in binary format and it can be used, as it is, as input for FATIGA and GRIETA programs.

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REGISTRO DE REVISIONES/REVISIONS RECORD

Revisión	Motivo de Modificación/Change reason	Realiz./Prep.	Compr./Check.	Aprobado/App.
Fecha/Date	Capítulos,Secciones,Hojas afectadas/Chapters,Sections,Sheets affected	Firma/Sign	Firma/Sign	Firma/Sign
A	Primera edición / First issue	Darío Fernández	Lucas Claramonte	Darío Fernández
July 2008				
B	New .aux output, additional information in summary output and fix omission error for a continuous stress sequence	Eduardo Sánchez	Gustavo Baños	---
Febrary 2023	Pages 6,7, 9, 10, 13, 16-18, 22, 28 and 29			

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1.- Technical sheet of the program

Technical Note **NT-T-ADP-08005**

PROGRAM	DATE
PROCESASEC	2008

OBJECTIVE

This program processes a flight-by-flight, cycle-by-cycle random stress sequence to be used for further applications.

LANGUAGE	COMPUTER
Fortran 77	Sun Solaris
NUMBER OF LINES (APROX.)	EXECUTION MODE
2 000	Interactive
AUTHOR	VERSION
Dario Fernández Villalba	2.0

ADDITIONAL DOCUMENTS

Technical Note: NT–T–ADP–08006.
Theoretical Manual of the PROCESASEC Program, Version 2.

Technical Note: NT–T–ADP–08007.
Programmer’s Manual of the PROCESASEC Program, Version 2.

2.- Introduction

The purpose of this Technical Report is to define the way to use the PROCESASEC program.

This program processes a flight-by-flight, cycle-by-cycle random stress sequence. The input sequence can be provided in two different ways:

- Sequence of discrete flights.
- Continuous stress recording.

The processed sequence will be generated with the following particularities:

- The stress sequence is filtered, removing all intermediate stress states.
- The sequence can be modified by applying a factor on stresses and adding a constant stress level.
- Small cycles can be removed from the final sequence by defining an appropriate omission level.
- A Rainflow counting method can be applied on the sequence.
- A block spectrum is also generated from the final stress sequence.

The final stress sequence is written in binary format and it can be used, as it is, as input for FATIGA and GRIETA programs.

Differences Between versions:

- Version 1.0:

  Initial version developed in Fortran 77.

  In parallel, a Windows application has been developed as Graphic User Interface GUI for the program. This application uses the same input file and provides the same stress sequence and summary. The process is performed by calling to the Dynamic Link Library DLL, compiled from the same source code as the official Fortran version.

- Version 1.1:

  Some irrelevant changes are included in the Fortran version and the Graphic User Interface is improved.

- Version 1.2:

  Small changes in the procedure to filter and depurate the sequence are included to avoid the loss of some relevant stress level at the beginning or at the end of the sequence.

  The Graphic User Interface is improved and adapted to the new platform Microsoft .NET Framework.

- Version 1.3:

  It is solved the error produced for input sequences, given in terms of continuous stress recording, with a number of stress states exceeding the program capability

- Version 1.4:

  It is solved the error produced when attempting to apply the Rainflow on a sequence having a unique cycle. In such case, the Rainflow is not applied.

- Version 2.0:

  Now it is possible to apply a factor, positive or negative, on all stresses in the sequence and to add a constant stress level, as follows:

  $$\sigma = \sigma_{add} + \text{Factor} \cdot \sigma_o$$

- Version 2.1:

  It is solved the error produced when a more than 2 million stress states sequence was used.

- Version 2.2:

  It is added a previous filter and omission in continuous stress sequence to avoid memory problems.

- Version 2.3:

  Software code is updated to new Intel Parallel Studio XE 2013 Fortran environment.

- Version 2.4:

  It is added new enter point to DLL in order to make possible the tool usage without an input file.

  It is also added the Linux version.

- Version 2.5:

  It is increased the memory capability in order to process greater continuous stress sequences.

  The Graphic User Interface is improved.

  Software code is updated to Visual studio 2013 environment.

- Version 2.6:

  It is reduced the memory capability given in previous version to solve problems found.

- Version 2.7:

  Echo for discrete flights sequence input.

- Version 2.8:

  It is added a new .aux output file in continuous stress sequence mode showing the cycles and the input line for peak and valley for each of them.

  It is solved the error produce because of the filter and omission included in version 2.4 for a continuous stress sequence starting in decreasing stress point – one cycle was omitted.

  It is added in summary output file more information about block generation and valley-peak coefficients.

Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$procesasec
```

The program asks for the language to use, the parameters defining the process and the name of the input and output files, as shown in the following examples for a discrete sequence and continuous sequence respectively:

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2

Type of sequence:
=================

  1 ... Sequence of discrete flights

  2 ... Continuous stress recording

Select the type of sequence: 1

Type of Rainflow:
=================

  0 ... Without Rainflow

  1 ... Local Rainflow

  2 ... Local Rainflow + GAG

  3 ... Full Rainflow

  4 ... Full Rainflow rearranged

Select the type of rainflow to apply: 4

Omission level: 10.0

Additional stress: 10.0

Factor on stresses: 1.15
```

```
Name of the input file:  Example.dat

Name of the output file for the sequence: Example.cbc

Name of the output file for the summary: Example.sum
```

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2

Type of sequence:
=================

  1 ... Sequence of discrete flights

  2 ... Continuous stress recording

Select the type of sequence: 2

Type of Rainflow:
=================

  0 ... Without Rainflow

  1 ... Local Rainflow

  2 ... Local Rainflow + GAG

  3 ... Full Rainflow

  4 ... Full Rainflow rearranged

Select the type of rainflow to apply: 4

Omission level:  0.0

Additional stress:  0.0

Factor on stresses: 1.15
```

```
Name of the input file:  Example.dat

Name of the output file for the sequence: Example.cbc

Name of the output file for the summary: Example.sum
```

```
Extra sequence file .aux?
=========================

  0 ... No

  1 ... Yes

Select option: 1
```

The CPU time for calculation depends on the size of the input sequence, being a few seconds as much.

It is possible to execute the program with a single command, including the input and output filenames, process data and the language as parameters. The above examples can be executed as follows for discrete and continuous sequences:

```
$procesasec Example.dat Example.cbc Example.sum 1 4 10.0 10.0 1.15 2

$procesasec Example.dat Example.cbc Example.sum 2 4  0.0  0.0 1.15 2 1
```

The required parameters to be used in the above command, must be included in the following order:

- Input sequence filename
- Output sequence filename
- Summary filename
- Type of input sequence (1 or 2)
- Type of Rainflow counting method (0 to 4)
- Omission level
- Additional stress
- Factor on stresses, positive or negative, but not zero
- Language (1 or 2)
- Extra .aux sequence file in case of continuous stress sequence (0 or 1)

3.- Input sequence file

This file contains the input sequence to be processed by the program. The arrangement of this file depends on the type of sequence.

3.1 Sequence of discrete flights

In this case, the input stress sequence is defined as a succession of flights, each one of them containing several stress states. For this kind of sequence, the organisation of the input file is the following:

```
Nflt
  Nst(1)                                  }
    Stress(1,1)                           }
    Stress(1,2)                           }
    ...........                           }  Flight 1
    ...........                           }
    Stress(1,Nst(1))                      }
  Nst(2)                                  }
    Stress(2,1)                           }
    Stress(2,2)                           }
    ...........                           }  Flight 2
    ...........                           }
    Stress(2,Nst(2))                      }
  ..................
  ..................
  ..................
  ..................
  Nst(Nflt)                               }
    Stress(Nflt,1)                        }
    Stress(Nflt,2)                        }
    ...........                           }  Flight Nflt
    ...........                           }
    Stress(Nflt,Nst(Nflt))                }
```

The meaning of each one of the above parameters is the following:

- Nflt: Is the number of flights in the sequence.
- Nst(i): Is the number of stress states contained in the flight “i”.
- Stress(i,j): Is the stress for the state “j” in the flight “i”.

Stresses contained in the same flight can be written grouped in the same line.

3.2 Continuous stress recording

In this case, the input stress sequence is defined as a succession of stress states, without any indication of beginning or end of flight. For this kind of sequence, the organisation of the input file is the following:

```
Code(1)   Stress(1)
Code(1)   Stress(2)
.......   .........
.......   .........
Code(n)   Stress(n)
```

The meaning of each one of the above parameters is the following:

- Code(i): Is a number that can be used to identify the origin of the stress state. Nevertheless the program ignores this value
- Stress(i): Is the stress in the state "i" of the sequence.

4.- Output results

The PROCESASEC program creates two files for both discrete and continuum sequence modes, one of them is not printable, with the flight-by-flight, cycle-by-cycle random stress sequence, and the other is a printable file, containing the summary of the processed sequence. For continuous stress sequence mode an extra file could be obtained showing the output stress sequence file in a printable form.

– Output stress sequence file.

– Output summary file.

– Continuous extra output stress sequence file.

4.1 Output stress sequence file

This file contains the flight by flight, cycle by cycle stress sequence and it is written in binary form, compatible with other programs, like GRIETA (input file), FATIGA (input file) or SECUENCIA (output file).

The first record has the number of flights in the sequence, and the subsequent records have a complete flight each one.

This file can be read by means of the following sub program (written in Fortran):

```
****************
* DECLARATIONS *
****************
      INTEGER*4 NFL,NCYC(xx),IFL,ICYC
      REAL*4 SMIN(xx,yy),SMAX(xx,yy)


****************
* FILE OPENING *
****************
      OPEN (1,FILE='filename',STATUS='OLD',FORM='UNFORMATTED')

***************************
* TOTAL NUMBER OF FLIGHTS *
***************************
      READ (1) NFL

*******************
* LOOP OF FLIGHTS *
*******************
      DO IFL=1,NFL

***********************
* READING OF A FLIGHT *
***********************
      READ (1) IFL,NCYC(IFL),(SMIN(IFL,ICYC),SMAX(IFL,ICYC),ICYC=1,NCYC(I))

**************************
* END OF LOOP OF FLIGHTS *
**************************
      END DO
```

```
***************
* FILE CLOSING *
***************
      CLOSE(1)
```

The meaning of each one of the above variables is the following:

NFL.................................. Is the total number of flights in the sequence.

NCYC(IFL) .... Is the number of cycles in the flight number “IFL”.

SMIN(IFL,ICYC)............ Is the minimum of the cycle number “ICYC” in the flight number “IFL”.

SMAX(IFL,ICYC) .... Is the maximum of the cycle number “ICYC” in the flight number “IFL”.

IFL and ICYC.................. Are loop variables.

When the input stress sequence is defined as a sequence of discrete flights, the number of flights in the output sequence “NFL” is the same as for the input sequence.

If the input stress sequence is defined as a continuous recording, the number of flights in the output sequence will be NFL = 1.

4.2 Output summary file

This file includes the following information:

- **Block spectrum:**

  The block spectrum contains all cycles of the output stress sequence, grouped in blocks of cycles. It can be used as input spectrum for the program GRIETA. It has one line for each block, in the following way:

Maximum stress	Minimum stress	Number of cycles of the block
Maximum stress	Minimum stress	Number of cycles of the block
. . . .	. . . .	. . . .
. . . .	. . . .	. . . .
Maximum stress	Minimum stress	Number of cycles of the block

- **Summary of the random stress sequence:**

  This section of the file contains the following data:

  – Table of the peaks distribution, in the form of an interval of peak stresses versus the frequency.

  – Table of the amplitude distribution, given in the same way.

  – Table of the valley-peak ratio distribution, also given in the same way.

– A summary showing: Maximum peak
Minimum valley
Maximum range
Number of total cycles
Number of total flights
Number of cycles per flight

Examples of the output summary file are provided in chapter 7.

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. The error messages that might appear are listed on next page:

```
ERROR: The input file cannot be opened
ERROR: The output file for the sequence cannot be opened
ERROR: The output file for the summary cannot be opened
ERROR: Invalid type of sequence
ERROR: Invalid type of Rainflow
ERROR: Invalid omission level
ERROR: Invalid additional stress
ERROR: Invalid factor on stresses
ERROR: Bad input data in input sequence file
ERROR: End of file while reading the input sequence file
```

<u>Error messages. Spanish output:</u>

```
ERROR: No se puede abrir el fichero de entrada
ERROR: No se puede abrir el fichero de salida para la secuencia
ERROR: No se puede abrir el fichero de salida para el sumario
ERROR: Tipo de secuencia incorrecto
ERROR: Tipo de "Rainflow" incorrecto
ERROR: Nivel de omisión incorrecto
ERROR: Esfuerzo adicional incorrecto
ERROR: Factor sobre los esfuerzos incorrecto
ERROR: Datos erróneos en el fichero de la secuencia de entrada
ERROR: Fin de fichero mientras se leía la secuencia de entrada
```

4.3 Continuous extra output stress sequence file

This file contains the flight by flight, cycle by cycle stress sequence and it is written in legible form. It is only available in continuous sequence mode and it also contains the input file line for the peak and valley stress values for each cycle.

5.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The application can be executed by "double click" in the file **ProcesaSec.exe**, located in the PC's server, or by means a **link** in the "desktop" or in the "start menu".

The input data must be typed in the application form. This form, after typing the data corresponding to the example 1 (see section 7.1), has the following appearance:

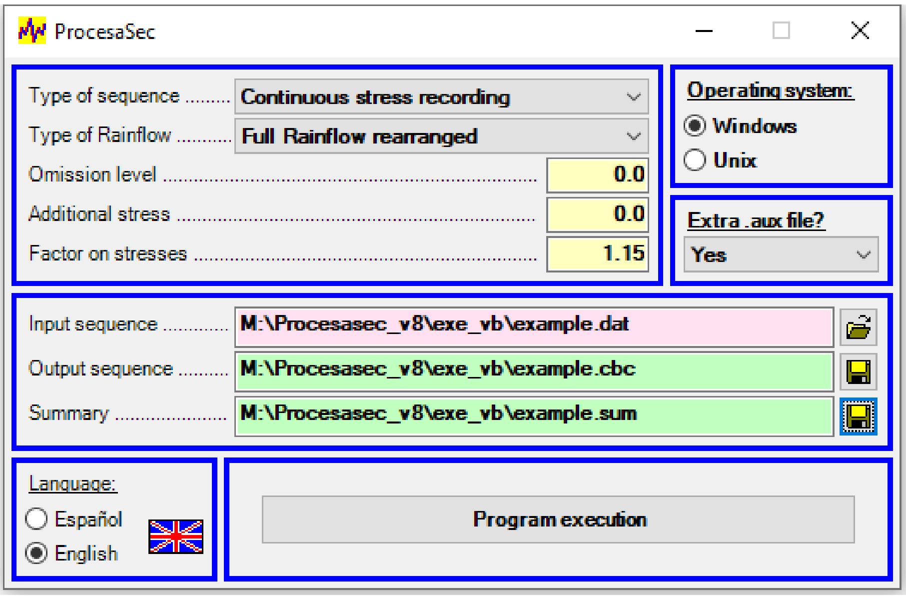

Figure 1: Input data form.

In the above form, the following data have been specified:

- The input sequence is provided as a continuous stress sequence.
- A "full Rainflow rearranged" will be applied on the stress sequence, with a constant omission level of 0 N/mm^2 (assuming that the input sequence is also expressed in N/mm^2).
- Before the above process, a factor of 1.15 will be applied on the original sequence and a constant stress level of 0.0 will be added.
- The output stress sequence will be generated as a binary file for Microsoft Windows operating system. Therefore it cannot be used as input for applications installed in Unix.
- The extra .aux output stress sequence will be generated as a legible file.

Note that it is also possible to generate the sequence as a binary file for a computer with Unix operating system by selecting the Unix option in the above form. In such a case, the sequence could be used as input for applications installed in Unix machines, but not for Windows.

The program can be then executed by clicking the button “Program execution”. The application starts to solve the problems showing the following form at the end:

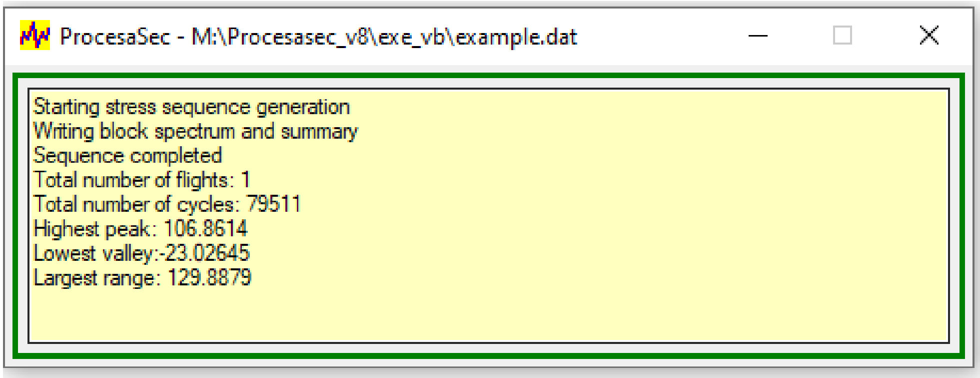

Figure 2: Tracking and results form.

The output results are written in the output files ***Example.cbc*** (binary file with the random stress sequence), ***Example.aux*** (legible form of the stress sequence) and ***Example.sum*** (printable file with the sample, the block spectrum and the summary).

6.- Examples

Two examples of how to use the program are given below. Each one of them corresponds to a different input sequence format.

6.1 Example 1

- Input stress sequence:

  It is a sequence of five discrete flights. All them are equal, with the following sequence of stresses each one, expressed in N/mm^2:

  ```
  60.0 40.0 20.0 50.0 80.0 60.0 40.0 70.0 100.0 45.0 -10.0 -5.0 0.0
  ```

- Output stress sequence:

  The above stresses will be multiplied by a factor of 1.15 and, after that, a constant stress level of 10 N/mm^2 will be added.

  It will be generated applying a “Full Rainflow Rearranged” counting method with an omission level of 10 N/mm^2 in range.

The input sequence as well as the output summary file are provided on next pages:

Input stress sequence file:

```
5
  13
    60.0    40.0    20.0    50.0    80.0
    60.0    40.0    70.0   100.0    45.0
   -10.0    -5.0     0.0
  13
    60.0    40.0    20.0    50.0    80.0
    60.0    40.0    70.0   100.0    45.0
   -10.0    -5.0     0.0
  13
    60.0    40.0    20.0    50.0    80.0
    60.0    40.0    70.0   100.0    45.0
   -10.0    -5.0     0.0
  13
    60.0    40.0    20.0    50.0    80.0
    60.0    40.0    70.0   100.0    45.0
   -10.0    -5.0     0.0
  13
    60.0    40.0    20.0    50.0    80.0
    60.0    40.0    70.0   100.0    45.0
   -10.0    -5.0     0.0
```

Output summary file. Block spectrum:

```
****** B L O C K   S P E C T R U M ******

       Smax       Smin    Cycles
       ----       ----    ------
      77.50      34.88         5
     102.50      56.38         5
     127.50      -6.38         5
```

Output summary file. Sequence summary:

```
************ O V E R A L L   S E Q U E N C E   S U M M A R Y *************

 Peak distribution              Range distribution              Valley/Peak distribution
 =================              ==================              ========================
  0.00   5.00       0             0.00   5.00       0                  <    -1.40       0
  5.00  10.00       0             5.00  10.00       0              -1.40   -1.30       0
 10.00  15.00       0            10.00  15.00       0              -1.30   -1.20       0
 15.00  20.00       0            15.00  20.00       0              -1.20   -1.10       0
 20.00  25.00       0            20.00  25.00       0              -1.10   -1.00       0
 25.00  30.00       0            25.00  30.00       0              -1.00   -0.90       0
 30.00  35.00       0            30.00  35.00       0              -0.90   -0.80       0
 35.00  40.00       0            35.00  40.00       0              -0.80   -0.70       0
 40.00  45.00       0            40.00  45.00       0              -0.70   -0.60       0
 45.00  50.00       0            45.00  50.00      10              -0.60   -0.50       0
 50.00  55.00       0            50.00  55.00       0              -0.50   -0.40       0
 55.00  60.00       0            55.00  60.00       0              -0.40   -0.30       0
 60.00  65.00       0            60.00  65.00       0              -0.30   -0.20       0
 65.00  70.00       0            65.00  70.00       0              -0.20   -0.10       0
 70.00  75.00       0            70.00  75.00       0              -0.10    0.00       5
 75.00  80.00       5            75.00  80.00       0               0.00    0.10       0
 80.00  85.00       0            80.00  85.00       0               0.10    0.20       0
 85.00  90.00       0            85.00  90.00       0               0.20    0.30       0
 90.00  95.00       0            90.00  95.00       0               0.30    0.40       0
 95.00 100.00       0            95.00 100.00       0               0.40    0.50       5
100.00 105.00       5           100.00 105.00       0               0.50    0.60       5
105.00 110.00       0           105.00 110.00       0               0.60    0.70       0
110.00 115.00       0           110.00 115.00       0               0.70    0.80       0
115.00 120.00       0           115.00 120.00       0               0.80    0.90       0
120.00 125.00       0           120.00 125.00       0               0.90    1.00       0
125.00 130.00       5           125.00 130.00       5               1.00    1.10       0
130.00 135.00       0           130.00 135.00       0               1.10    1.20       0
135.00 140.00       0           135.00 140.00       0               1.20    1.30       0
140.00 145.00       0           140.00 145.00       0               1.30    1.40       0
145.00 150.00       0           145.00 150.00       0               1.40     <          0
150.00 155.00       0           150.00 155.00       0
155.00 160.00       0           155.00 160.00       0          Maximum peak   =     125.00
160.00 165.00       0           160.00 165.00       0          Minimum valley =      -1.50
165.00 170.00       0           165.00 170.00       0          Maximum range  =     126.50
170.00 175.00       0           170.00 175.00       0          Maximum R      =       0.55
175.00 180.00       0           175.00 180.00       0          Minimum R      =      -0.01
180.00 185.00       0           180.00 185.00       0
185.00 190.00       0           185.00 190.00       0          Cycles         :         15
190.00 195.00       0           190.00 195.00       0          Flights        :          5
195.00 200.00       0           195.00 200.00       0          Cycles/Flight  :       3.00
```

Example 2

- Input stress sequence:

  It is a continuous stress recording with 263086 stress states.

- Output stress sequence:

  All stresses in the input sequence will be multiplied by a factor of 1.15. No constant stress level will be added.

  It will be generated applying a “Full Rainflow Rearranged” counting method without any omission level.

The input sequence as well as the output summary file are provided on next pages:

Input stress sequence file:

```
1048     -0.092
1048     10.690
1048     29.140
1048     28.010
1048     10.690
1048     21.472
1048     10.690
1008      4.555
1008     10.690
1019     14.079
1019     10.690
1019     29.140
1019     28.010
1019     10.690
1019      7.301
1019     10.690
1003      4.555
1003     10.690
1035     17.159
1035     10.690
1035     40.830
1035     45.430
1035     10.690
1035      4.221
1035     10.690
1003      4.555
1003     10.690
1036      4.221
1036     10.690
1036     40.830
1036     45.430
1036     10.690
1036     17.159
1036     10.690
1003      4.555
1003     10.690
1009     29.140
1009     28.010
1009     10.690
1003      4.555
1003     10.690
1019      7.301
1019     10.690
1019     29.140
1019     28.010
1019     10.690
1019     14.079
1019     10.690
1003      4.555
....     ......
....     ......
....     ......
....     ......
```

Output Summary file. Block spectrum:

```
****** B L O C K   S P E C T R U M ******

           Smax      Smin    Cycles
           ----      ----    ------
          12.50     -0.62        68
          12.50      4.38       246
          12.50      5.62      1437
          12.50      6.88       314
          12.50      8.12      7838
          17.50     -0.88        39
          17.50      4.38        41
          17.50      7.88         5
          17.50      9.62        42
          17.50     11.38      1404
          17.50     13.12      6904
          22.50     -1.12         1
          22.50      3.38         6
          22.50      7.88         1
          22.50     10.12       497
          27.50      6.88        52
          27.50      9.62         1
          27.50     12.38         1
          27.50     15.12       220
          27.50     17.88      2729
          27.50     20.62     10762
          32.50    -14.63         5
          32.50    -11.38         1
          32.50     -1.62        14
          32.50      4.88      4470
          32.50      8.12       472
          32.50     11.38        55
          32.50     14.62        16
          32.50     21.12       113
          32.50     27.62      9630
          32.50     30.88       254
          37.50     -1.88         3
          37.50      5.62        93
          37.50      9.38         1
          37.50     13.12        10
          37.50     20.62        12
          37.50     24.38      1927
          42.50    -14.88         5
          42.50     -2.12        11
          42.50      6.38        14
          47.50    -16.63         1
          47.50    -11.88        12
          47.50     -2.38         9
          47.50      2.38         3
          47.50      7.13       225
          47.50     16.62         3
          47.50     21.38        12
          47.50     26.12       699
          47.50     40.38         6
          47.50     45.12        52
```

<u>Output Summary file. Block spectrum:</u> (cont.)

```
****** B L O C K   S P E C T R U M ******

          Smax       Smin     Cycles
          ----       ----     ------
         52.50     -18.38          1
         52.50     -13.12         25
         52.50      -2.62        155
         52.50       2.62        612
         52.50       7.88       2557
         52.50      13.12         30
         52.50      18.38          1
         52.50      44.62          8
         52.50      49.88         12
         57.50     -14.38         83
         57.50      -2.88         11
         57.50       2.88         31
         57.50       8.62         24
         57.50      14.38        731
         57.50      20.12      10418
         57.50      54.62         72
         62.50     -21.88          2
         62.50     -15.62        173
         62.50      -9.38          6
         62.50      -3.12        239
         62.50       3.12       1817
         62.50       9.38          7
         62.50      15.62          6
         67.50     -16.88         19
         67.50     -10.12          1
         67.50      10.12          3
         67.50      16.88         75
         72.50      10.88          2
         72.50      18.12         17
         77.50     -11.62        107
         77.50       3.88        154
         77.50      11.62        460
         77.50      19.38       6323
         77.50      73.62        182
         82.50     -12.38          4
         82.50      -4.12          7
         82.50       4.12         45
         82.50      12.38         27
         82.50      20.62        268
         87.50     -21.88          1
         87.50     -13.13        291
         87.50      -4.38         10
         87.50       4.38        239
         87.50      13.13       2147
         87.50      21.88         26
         87.50      83.12        100
         92.50     -23.12          1
         92.50     -13.88        204
         92.50      -4.62         15
         92.50       4.62        273
```

<u>Output Summary file. Block spectrum:</u> (cont.)

```
****** B L O C K   S P E C T R U M ******

           Smax       Smin     Cycles
           ----       ----     ------
          92.50      13.88        345
          92.50      23.12         13
          92.50      87.88         35
          97.50     -14.63        162
          97.50      -4.88          2
          97.50       4.88         51
          97.50      14.63        115
          97.50      24.38          1
         102.50     -25.62          5
         102.50     -15.38         14
         102.50       5.12          4
         102.50      15.38          3
         107.50     -26.88          1
         107.50     -16.12          2
```

Output summary file. Sequence summary:

```
************ O V E R A L L   S E Q U E N C E   S U M M A R Y *************

 Peak distribution          Range distribution          Valley/Peak distribution
 =================          ==================          ========================
  0.00   5.00       0        0.00   5.00  15433              <   -1.40       0
  5.00  10.00       0        5.00  10.00  26594          -1.40   -1.30       0
 10.00  15.00    9903       10.00  15.00   2871          -1.30   -1.20       0
 15.00  20.00    8435       15.00  20.00    827          -1.20   -1.10       0
 20.00  25.00     505       20.00  25.00     66          -1.10   -1.00       0
 25.00  30.00   13765       25.00  30.00   4955          -1.00   -0.90       0
 30.00  35.00   15030       30.00  35.00    114          -0.90   -0.80       0
 35.00  40.00    2046       35.00  40.00  10408          -0.80   -0.70       0
 40.00  45.00      30       40.00  45.00   1138          -0.70   -0.60       0
 45.00  50.00    1022       45.00  50.00   3160          -0.60   -0.50       0
 50.00  55.00    3401       50.00  55.00    226          -0.50   -0.40       5
 55.00  60.00   11370       55.00  60.00   2120          -0.40   -0.30      10
 60.00  65.00    2250       60.00  65.00   6885          -0.30   -0.20     320
 65.00  70.00      98       65.00  70.00    268          -0.20   -0.10     791
 70.00  75.00      19       70.00  75.00   2554          -0.10    0.00     584
 75.00  80.00    7226       75.00  80.00    366           0.00    0.10    3229
 80.00  85.00     351       80.00  85.00    583           0.10    0.20   10498
 85.00  90.00    2814       85.00  90.00     96           0.20    0.30    8056
 90.00  95.00     886       90.00  95.00    164           0.30    0.40   10735
 95.00 100.00     331       95.00 100.00      4           0.40    0.50    1968
100.00 105.00      26      100.00 105.00    459           0.50    0.60    1287
105.00 110.00       3      105.00 110.00    141           0.60    0.70   14011
110.00 115.00       0      110.00 115.00     53           0.70    0.80   17666
115.00 120.00       0      115.00 120.00     15           0.80    0.90    9644
120.00 125.00       0      120.00 125.00      8           0.90    1.00     707
125.00 130.00       0      125.00 130.00      3           1.00    1.10       0
130.00 135.00       0      130.00 135.00      0           1.10    1.20       0
135.00 140.00       0      135.00 140.00      0           1.20    1.30       0
140.00 145.00       0      140.00 145.00      0           1.30    1.40       0
145.00 150.00       0      145.00 150.00      0           1.40     <         0
150.00 155.00       0      150.00 155.00      0
155.00 160.00       0      155.00 160.00      0      Maximum peak   =     106.86
160.00 165.00       0      160.00 165.00      0      Minimum valley =     -23.03
165.00 170.00       0      165.00 170.00      0      Maximum range  =     129.89
170.00 175.00       0      170.00 175.00      0      Maximum R      =       1.00
175.00 180.00       0      175.00 180.00      0      Minimum R      =      -0.41
180.00 185.00       0      180.00 185.00      0
185.00 190.00       0      185.00 190.00      0      Cycles         :     79511
190.00 195.00       0      190.00 195.00      0      Flights        :         1
195.00 200.00       0      195.00 200.00      0      Cycles/Flight  :  79511.00
```

Output Extra sequence file:

```
ProcesaSec 2.8      (c) Airbus Defence and Space, Feb-2023       24-Feb-2023
--------------------------------------------------------------------------------

    Smin       Smax    LCSmin    LCSmax
    5.24      33.51         8         3
   12.29      24.69         5         6
   12.29      16.19        11        10
    5.24      33.51        17        12
    8.40      12.29        15        16
   12.29      19.73        20        19
    4.85      52.24        24        22
    5.24      12.29        26        25
    4.85      12.29        28        27
    4.85      52.24        51        31
   12.29      19.73        32        33
    5.24      33.51        35        37
    8.40      12.29        42        41
    5.24      33.51        40        44
   12.29      16.19        46        47
    5.24      12.29        49        50
    4.85      33.51        60        53
   12.29      19.73        55        56
    5.24      12.29        58        59
   -0.11      61.38         1        63
   12.29      19.73        64        65
   12.29      16.19        70        69
    5.24      52.24        67        72
    8.40      12.29        74        75
   12.29      24.69        79        78
    5.24      33.51        76        80
    5.24      12.29        85        84
    5.24      52.24        94        90
    8.40      12.29        92        93
    8.40      12.29        96        95
    4.85      52.24       110        99
   12.29      16.19       100       101
   12.29      19.73       106       105
    5.24      33.51       103       107
    5.24      12.29       112       111
    8.40      12.29       114       113
    5.24      54.37       121       118
   12.29      19.73       124       123
    4.85      61.38       164       126
    5.24      12.29       130       129
    4.85      12.29       132       131
    4.85      52.24       128       135
   12.29      19.73       136       137
   12.29      19.73       142       141
    5.24      33.51       139       143
  ……..       ……..      ……..      ……..
  ……..       ……..      ……..      ……..
```